EE 156 – CHAPTER 2

Fundamentals of Electrical Controls and PLC Systems

Relays and Contactors

2.1 INTRODUCTION AND FUNCTIONS

What are Relays and Contactors?

- **Electrically operated switches** that enable control of electrical circuits using **low-power control signals**
- Both perform switching functions but are designed for **different applications and power levels**
- Based on the **solenoid (electromagnetic) principle**

Basic Operating Principle

1. Current flows through the coil
2. Magnetic field is produced
3. Magnetic field attracts movable iron armature
4. Armature movement causes contacts to **open or close**

2.2 RELAYS

Definition

A relay is an **electrically operated switching device** that uses an electromagnetic coil to open or close one or more contacts.

Functions of Relays

Function	Description
Electrical isolation	Separate control and power circuits
Remote switching	Control devices from a distance
Logic operations	Facilitate control circuit logic
Equipment protection	Fault detection and switching
Amplification	Small current controls larger current

Characteristics

- **Electromechanical switches** with auxiliary contacts
- Usually **one coil** but may have multiple contacts
- Current ratings: **1 to 10 amperes** (typical)

- Contacts are **small** and intended for **control applications**

2.3 CONTACTORS

Definition

A **heavy-duty electromagnetic switching device** designed specifically for controlling **high-current loads**.

Functions of Contactors

- Start and stop electric motors
- Control power delivery to industrial loads
- Provide safe switching of high-current circuits
- Enable remote and automatic control
- Facilitate motor control and automation

Key Difference from Relays

- **Contactors** handle **high power** (motors, heaters, lighting)
- **Relays** handle **low power** (control circuits, signal processing)

2.4 CONSTRUCTION AND OPERATION

Construction Differences: AC vs DC

Feature	DC Relay/Contactor	AC Relay/Contactor
Core material	Solid core	Laminated core
Reason	–	Reduces eddy current losses
Shading coil	Not required	Required (prevents chattering)

Why Laminated Cores for AC?

- AC causes **continuously changing electromagnetic field**
- Leads to **core losses** (hysteresis and eddy currents)
- Laminations **reduce eddy current losses**

Terminals of a Relay

Terminal	Symbol	Description
Coil/Control input	A1, A2	Input terminals to activate/deactivate relay
Common (COM)	C	Output terminal connected to load circuit
Normally Open (NO)	NO	Open when relay not active; closes when energized
Normally Closed (NC)	NC	Closed when relay not active; opens when energized

Internal Construction

```
             +-----------------+
             |   Stationary    |
             |   Contact (NC)  |
             |        |        |
Coil --------+--[Iron Core]----+---------------
             |        |        |
             |   [Armature]    |
             |    (Movable)    |
             |        |        |
             |   Stationary    |
             |   Contact (NO)  |
             +-----------------+
                  Spring
             (Returns to NC)
```

Operation Sequence

1. Energized (Active):

- Coil creates magnetic field

- Armature attracted to iron core
- Movable contact **breaks away** from NC
- Makes connection with **NO**
- COM connects to **NO**

2. De-energized (Inactive):

- No magnetic field
- Spring returns armature
- Movable contact connects to **NC**
- COM connects to **NC**

Shading Coil

- **Required for AC operation**
- Prevents **contact chattering**
- Creates phase-shifted magnetic field

2.5 TYPES OF RELAY CONSTRUCTION

Two Basic Methods

1. Clapper Type

- Uses **one movable contact** to make connection with a stationary contact
- Simple construction
- Common in many applications

2. Bridge Type

- Uses a **movable contact** to make connection **between two stationary contacts**
- More complex
- Provides better contact pressure

2.6 RELAY CLASSIFICATION BY POLES AND THROW

Key Definitions

Term	Meaning
Pole	Number of switches inside a relay
Throw	Number of circuits controlled per pole

Types of Relays Based on Poles and Throw

1. SPST Relay (Single Pole Single Throw)

Feature	Description
Poles	1 (controls one circuit)
Throw	1 (one conducting position)
States	Either OPEN or CLOSED
Circuit	On/Off control

Example Application: Simple ON/OFF switching

```
        +------+
--------|  SPST |--------
        +------+
```

2. SPDT Relay (Single Pole Double Throw)

Feature	Description
Poles	1 (controls one circuit at a time)
Throw	2 (two conducting positions)
States	One circuit closed, other open (and vice versa)

Terminals: COM, NO, NC

```
            +-- NO
------------|
            +-- NC
```

Example Applications:

- Select between two circuits
- Changeover switching
- Polarity reversal

3. DPST Relay (Double Pole Single Throw)

Feature	Description

Poles	2 (controls two isolated circuits)
Throw	1 (each pole has one conducting position)
Action	Switches two circuits simultaneously

Example Applications:

- Simultaneous switching of two circuits
- Both circuits ON or both OFF together
- Double line switching

```
------------+     +----------
------------|     |----------
            |     |
------------|     |----------
------------+     +----------
```

4. DPDT Relay (Double Pole Double Throw)

Feature	Description
Poles	2 (controls two circuits)
Throw	2 (each pole can conduct in two positions)
Interpretation	Two SPDT relays, switching simultaneously

Example Applications:

- Motor forward/reverse control
- Switching two circuits simultaneously
- Complex control systems

```
------------+     +-- NO1
            |     +-- NC1
------------|     |
            |     +-- NO2
------------+     +-- NC2
```

Note: A relay can have **as many as 12 poles**!

2.7 TYPES OF RELAYS BY APPLICATION

2.7.1 Control Relays

Definition: Electromechanical relays specifically designed for **control and signaling** purposes.

Features:

- Multiple sets of contacts
- Switch multiple circuits simultaneously
- Contacts can be NO, NC, or both
- Handle **both AC and DC voltages**

Applications:

- Industrial control circuits
- PLC interfacing
- Logic control

2.7.2 Solid-State Relays (SSRs)

Definition: A **non-contact control device** using **semiconductors** to complete ON/OFF action.

AC Solid-State Relays

- Use a **triac** (bidirectional device)
- Permits current flow in either direction
- When energized, relay contact closes
- Supplies power to triac gate
- Triac connects load to line

DC Solid-State Relays

- Use a **transistor** instead of triac
- Connects DC load to line

Isolation Methods

Method	Description
Optical	Uses LED and photo-triac
Transformer	Magnetic isolation
Reed	Reed switch isolation

Optical Isolation (Optoisolation)

- Most common method
- Light from **LED** energizes a **photo triac**
- Provides **electrical isolation** between control and load sides
- Prevents **electrical noise** transfer

Advantages of SSRs

Advantage	Explanation
No moving parts	No contact wear
Long life	Higher reliability
Fast switching	No mechanical delay
Isolation	Control side isolated from load side
No contact bounce	Clean switching
Silent operation	No mechanical noise

Applications:

- PLC output modules

- Computer interfacing
- High-speed switching
- Environments with vibration

2.7.3 Timing Relays

Definition: Combination of an **electromechanical output relay** and a **control circuit** that opens or closes contacts after a **preselected time interval**.

Two Main Classifications

Type	Symbol	Meaning
On-Delay (DOE)	Delay On Energize	Contacts change after time delay when coil energized
Off-Delay (DODE)	Delay On De-Energize	Contacts change after time delay when coil de-energized

Types of Timing Relays

Type	Operating Principle	Characteristics
Pneumatic Timer	Restricted airflow through orifice	Unaffected by temperature variations; Wide time range
Clock Timer	AC synchronous motor	High repeat accuracy; Easy adjustment
Motor-Driven Timer	Small motor with gear train	Definite on/off operations; Series of operations
Electronic Timer	Electronic circuitry (microcontrollers)	Programmable; Highly accurate
Capacitor Timer	Capacitor charging/discharging	High accuracy; Motor control

Pneumatic Timers

Operating Principle:

- Uses **piston and control valve**
- Powered by **compressed air**
- Air flow restricted through orifice
- Expands/contracts rubber bellows or diaphragms

Advantages:

1. Unaffected by temperature variations
2. Adjustable over wide time range
3. Good repeat accuracy
4. Variety of contact arrangements

Applications:

- Areas where electrical current is potentially harmful
- Explosive environments
- Industrial processes

Electronic Timer Relay

Features:

- Electronic circuitry for timing
- Uses microcontrollers or integrated circuits
- Digital displays
- Programmable
- Customizable settings

Applications:

- Industrial process control
- Lighting control systems
- Cooking/Exercise equipment
- Precision timing requirements

2.7.4 Overload Relays

Definition: Protection devices that:

1. Identify **excessive current flow**
2. Respond and interrupt current through motor or equipment

Purpose:

- **Safeguard motors** from damage
- Prevent operation beyond limits
- Part of motor starter

Types of Overload Relays

Type	Operating Principle	Best For
Thermal Overload	Heat detection	Ambient temperature stable areas
Magnetic Overload	Magnetic field	Extreme ambient temperature changes

Thermal Overload Relay

Operating Principle:

- **Heater in series** with motor
- Heat produced depends on **motor current**
- Trips faster in **warm areas**

Sub-Types:

1. **Solder Melting Type (Solder Pot)** – Solder melts at predetermined temperature; releases mechanism to open contacts
2. **Bimetal Strip Type** – Two metals with different expansion rates; bends when heated; opens contacts

Applications:

- Motor protection
- Transformer protection
- Heavy-duty electrical equipment

Magnetic Overload Relay

Operating Principle:

- Operates on **magnetic field** generated by current
- **Adjustable trip points**
- Not affected by **ambient temperature**

Sub-Types:

1. **Electronic Magnetic Overload** – Uses electronic sensing; more accurate; programmable
2. **Dashpot Magnetic Overload** – Uses oil dashpot for time delay; mechanical design; simple operation

Applications:

- Motor control centers
- Switchboards
- Areas with extreme temperature changes

2.7.5 Phase Failure Relays

Definition: Electrical protection device that detects **absence or imbalance of phases** in a three-phase electrical system.

Functions:

- Detect phase loss or phase asymmetry
- Prevent damage to motors and equipment
- Trigger alarms or preventive measures
- Trip contactors or circuit breakers

Operating Principle:

- Compares amplitudes and phases
- Activates when variation exceeds preset thresholds
- Opens circuit breakers or trips contactors

Applications:

- Three-phase motor protection
- Pump protection
- Compressor protection
- Industrial machinery

2.8 DIFFERENCES: RELAY vs CONTACTOR

Feature	Relay	Contactor
Power rating	Low (1-10A)	High (tens to hundreds of amps)
Primary use	Control circuits	Power circuits
Load type	Signal/Control	Motors, heaters, lighting
Contact size	Small	Large
Arc suppression	Minimal	Arc chutes, blowouts
Contact material	Silver alloy	Silver-cadmium oxide
Auxiliary contacts	Many	Some
Main contacts	Few (NO/NC)	Usually NO
Cost	Lower	Higher
Physical size	Smaller	Larger

2.9 STANDARD SYMBOLS

Relay/Contactor Symbols

Component	Symbol
Coil (General)	Rectangle with diagonal line or circle
Relay Coil	[coil] between two terminals
Contactor Coil	[coil] (often with IEC symbols)
NO Contact	Two parallel bars, open gap
NC Contact	Two parallel bars with a diagonal crossing line
Overload Heater	Zig-zag element in series with line

Overload Relay Symbols

```
--| |--    (Thermal overload)

--| |--    (Magnetic overload)
```

Timer Symbols

Type	Symbol

On-Delay (NO)	Contact symbol with a small arc – closes after delay on energize
On-Delay (NC)	Contact symbol with a small arc – opens after delay on energize
Off-Delay (NO)	Contact symbol with a filled arc – closes after delay on de-energize
Off-Delay (NC)	Contact symbol with a filled arc – opens after delay on de-energize

2.10 APPLICATIONS SUMMARY

Relay Applications

- Control panels
- PLC interfacing
- Signal processing
- Protection systems
- Logic operations
- Low-power switching

Contactor Applications

- Motor starting and stopping
- Lighting control (high-power)
- Heating systems
- Industrial machinery
- Power distribution
- HVAC systems

2.11 KEY TERMINOLOGY

Term	Definition
Coil	Electromagnetic winding that creates magnetic field
Armature	Movable iron piece attracted by magnetic field
Contacts	Switching elements (NO, NC)
Pole	Number of switches in a relay
Throw	Number of positions per pole
Auxiliary contacts	Contacts used for control/signaling
Main contacts	Contacts used for load carrying
Shading coil	Copper ring on AC relay to prevent chatter
Arc chute	Device to extinguish electrical arc
Overload	Current exceeding rated value
Phase	Circuit in multi-phase system

REVIEW QUESTIONS

1. Explain the functions and applications of relays and contactors in electrical control systems.
2. What is the main difference between a relay and a contactor?
3. Draw and explain the terminals of a relay.
4. Why is a shading coil necessary in AC relays?
5. Differentiate between SPST and SPDT relays.
6. Explain the principle of operation of a solid-state relay (SSR).
7. What is optical isolation and why is it important?
8. Explain the difference between on-delay and off-delay timers.
9. Describe the operating principle of a thermal overload relay.
10. What is a phase failure relay and how does it protect equipment?

End of Chapter 2 Notes